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Predicting Credit Default: A Hybrid Econometric and Machine Learning Approach

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Abstract

This paper examines credit default prediction using a hybrid approach that combines traditional econometric models with machine learning techniques. Using the UCI Taiwan credit card dataset of 30,000 observations, we employ Logit and Probit models to identify key determinants of credit default, and XGBoost to improve predictive accuracy. To address multicollinearity among some of the original variables, we engineer a refined feature set using first differences, utilization ratios, and a cumulative delinquency score. Marginal effects from the econometric models reveal that cumulative payment delinquency and recent payment trajectory are the dominant predictors of default, increasing default probability by 3.3 and 5.3 percentage points. An equal-weighted ensemble of Logit, Probit, and XGBoost achieves the highest predictive performance with an AUC of 0.757, outperforming standalone econometric models (AUC 0.739) and XGBoost alone (AUC 0.753). These results suggest that econometric and machine learning approaches are complementary; econometrics provides inference and interpretability while machine learning captures nonlinear interactions that improve prediction. Default recall improves from 27.5% under Logit alone to 31.2% under the ensemble, demonstrating that combining both paradigms yields superior credit risk assessment relative to either approach in isolation.

Keywords: Credit Default Prediction, Logit, Probit, XGBoost, Ensemble Learning, Feature Engineering, Marginal Effects, Binary Classification, Payment Delinquency, ROC-AUC

Contents

1 Introduction **4**

1.1 Research Objective and Contribution 4

1.2 Organization of the Paper 4

2 Literature Review **6**

3 Data **6**

3.1 Variables 6

3.2 Data Cleaning and Feature Engineering 7

3.3 Class Imbalance 8

3.4 Data Analysis 8

3.5 Descriptive Statistics 11

4 Empirical Approach **12**

4.1 Why OLS Is Inappropriate for Binary Dependent Variables 12

4.2 Logit and Probit Models 12

4.3 Marginal Effects 13

4.4 Feature Engineering 13

4.5 Machine Learning Benchmark 13

4.6 Model Evaluation 14

5 Results **14**

5.1 Baseline LPM 14

5.2 Logit and Probit Results 15

5.3 Marginal Effects 16

5.4 Machine Learning Benchmarks 17

5.5 Ensemble Model 18

5.6 Model Comparison 18

6 Conclusion **20**

6.1 Summary of Findings 20

6.2 Limitations 21

6.3 Directions for Future Research 21

7 Acknowledgements **22**

1 Introduction

Consumer credit default poses a significant risk to financial institutions, affecting lending decisions, capital requirements, and portfolio stability. The ability to accurately identify borrowers at risk of default is crucial to credit risk management and has motivated a substantial amount of empirical research spanning several decades. Early approaches relied on linear discriminant analysis and rule-based scoring, while more recent work has shifted toward limited dependent variable models and machine learning methods that accommodate the binary nature of default outcomes and nonlinear relationships among predictors.

Despite this progress, a tension persists between predictive accuracy and interpretability. Machine learning models such as XGBoost and Random Forests achieve strong out-of-sample performance but offer limited structural insight into the drivers of default. Econometric models such as Logit and Probit provide coefficient estimates and marginal effects that are directly interpretable and policy-actionable, but may sacrifice predictive power relative to their ML counterparts.

1.1 Research Objective and Contribution

This paper addresses this tension directly. Using individual-level credit card data from Yeh and Lien (2009), this paper estimates and compares a Linear Probability Model, Logit, Probit, Random Forest, and XGBoost on a common feature set. A weighted ensemble model is then constructed at the end by combining Logit, Probit, and XGBoost predictions using constrained optimization to maximize AUC on a test set. The paper makes three contributions. First, it provides a systematic comparison of econometric and ML estimators on a well-known benchmark dataset. Second, it demonstrates that behavioural variables, particularly cumulative payment delay and recent delinquency, dominate demographic and credit limit variables as predictors of default. Third, it shows that combining econometric and ML predictions via ensemble methods generates robust discriminatory power relative to any individual model.

1.2 Organization of the Paper

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data sources and variable construction. Section 4 presents the econometric and machine learning methodologies. Section 5 reports empirical results and model comparisons.

Section 6 concludes and discusses implications for future research.

2 Literature Review

The prediction of credit default has a long history in empirical finance. Altman (1968) set the foundation of quantitative credit scoring using discriminant analysis, establishing the foundation for binary classification in credit markets. Subsequent work introduced limited dependent variable models as a more robust alternative, with logistic regression shown to outperform discriminant analysis for consumer credit applications due to its fewer distributional assumptions (Wiginton, 1980). The dataset used in this paper was introduced by Yeh and Lien (2009), who applied regression and neural network methods to predict credit card default among Taiwanese borrowers, providing the benchmark against which this paper's results are compared.

More recently, ensemble tree methods have emerged as a high-performing predictive tools in credit risk, with Breiman (2001) demonstrating that Random Forests yield significant gains in predictive accuracy over single classifiers through bootstrap aggregation. This paper builds on these contributions by benchmarking traditional econometric estimators against current ML methods and constructing a weighted ensemble that combines both approaches.

3 Data

This study uses the UCI Taiwan credit card dataset, which contains 30,000 observations of credit card clients in Taiwan from April to September 2005. The dataset was sourced from the UCI Machine Learning Repository and was originally studied by Yeh and Lien (2009). The binary response variable is whether a client defaulted on their credit card payment in the following month (1 = default, 0 = no default).

3.1 Variables

The dataset contains 23 exogenous variables across 4 different categories:

- **Credit Limit:** The amount of given credit in New Taiwan dollars, including both individual and supplementary credit.
- **Demographic Variables:** Gender (male = 1, female = 2), education level (graduate school = 1, university = 2, high school = 3, other = 4), marital status (married = 1, single = 2,

other = 3), and age in years.

- **Payment Status (X6–X11):** Monthly repayment status from April to September 2005. Coded as -1 = payment made in full, 1 = one month delay, 2 = two months delay, up to 9 = nine or more months delay.
- **Bill Statement Amounts (X12–X17):** Amount of bill statement in New Taiwan dollars from April to September 2005.
- **Payment Amounts (X18–X23):** Amount of previous payment in New Taiwan dollars from April to September 2005.

3.2 Data Cleaning and Feature Engineering

Undocumented categories in `EDUCATION` (values 0, 5, and 6) and `MARRIAGE` (value 0) were recoded into the existing "other" category following standard practice. No missing values were present in the raw data.

Three variable transformations were applied before estimation. First, raw payment status variables exhibit high serial correlation across months; borrowers who miss one payment tend to miss subsequent payments. First differences are constructed to capture the *change* in delinquency behaviour rather than its level:

$$\texttt{PAY_DIFF}_t = \texttt{PAY}_t - \texttt{PAY}_{t+1}$$

Second, a utilization ratio is constructed to capture credit usage intensity, a standard variable in the credit scoring literature:

$$\texttt{Utilization} = \frac{\texttt{BILL_AMT1}}{\texttt{LIMIT_BAL}}$$

Third, cumulative delinquency is measured as the sum of all positive payment status values across the observation window:

$$\texttt{TOTAL_DELAY} = \sum_t \max(\texttt{PAY}_t,\ 0)$$

The final model feature set comprises `LIMIT_BAL`, `SEX`, `EDUCATION`, `MARRIAGE`, `AGE`, first-differenced payment status variables (`PAY_DIFF_1` through `PAY_DIFF_5`), `Utilization`, and `TOTAL_DELAY`.

3.3 Class Imbalance

The dataset exhibits class imbalance, with 77.9% of observations classified as non-default and 22.1% as default. To preserve this distribution across model training and evaluation, the dataset was split into 75% training set (22,500 observations) and a 25% test set (7,500 observations) using stratified sampling.

3.4 Data Analysis

Figure 1 presents the distribution of the dependent variable. The dataset exhibits clear class imbalance, with 77.9% of observations classified as non-default and 22.1% as default. This imbalance is addressed through stratified sampling during the train-test split to ensure the default rate is preserved across both sets.

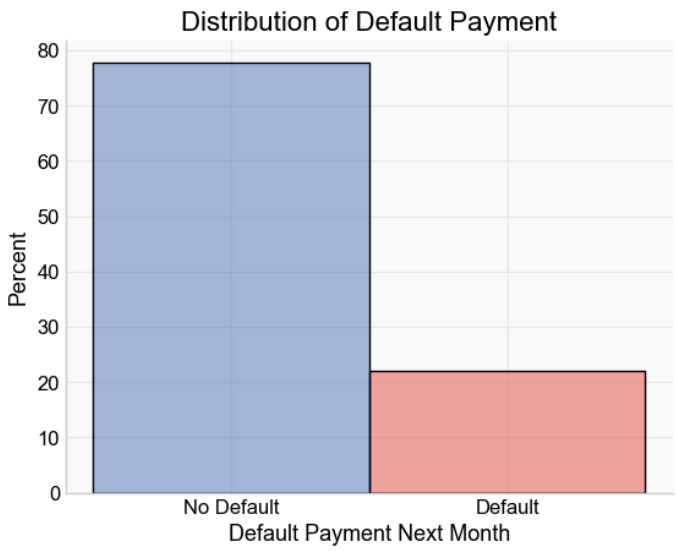

Figure 1: Distribution of Default Payment

Figure 2 presents the distribution of default status across major demographic characteristics. Female borrowers make up a substantial portion of the sample and show a higher absolute count of defaults, although this is partly driven by their larger presence in the dataset. Borrowers with a university education represent the highest proportion of defaults, which aligns with their predominant share in the sample. Married and single borrowers exhibit comparable default rates, indicating that marital status alone has limited predictive value. The age distribution shows that default risk is primarily concentrated among younger borrowers aged 20–40, and then declines steadily with age, consistent with lower credit experience and higher financial leverage typically observed among

younger individuals.

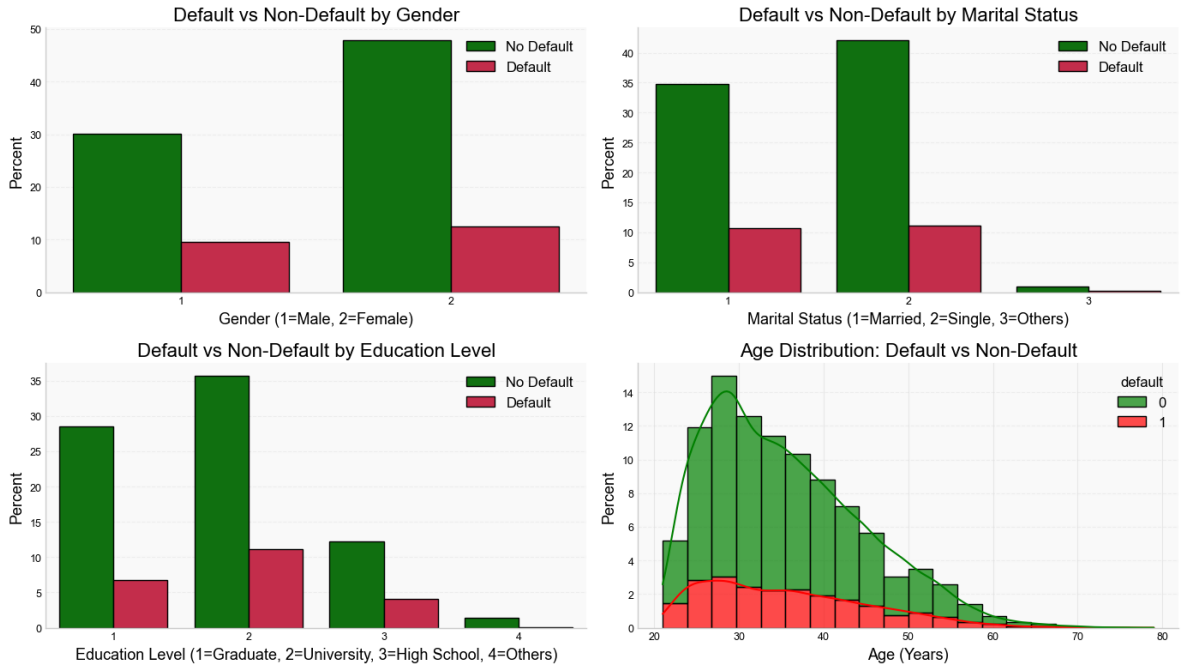

Figure 2: Default vs Non-Default by Demographic Characteristics

Figure 3 presents the correlation matrix for all raw features prior to transformation. Three clusters of multicollinearity are immediately apparent. First, the payment status variables (`PAY_0` through `PAY_6`) are strongly positively correlated, reflecting the persistence of delinquency behaviour. Second, bill statement amounts (`BILL_AMT1` through `BILL_AMT6`) exhibit high intercorrelation, as outstanding balances carry forward across months. Third, payment amounts show moderate correlation. These patterns directly motivate the feature engineering described above.

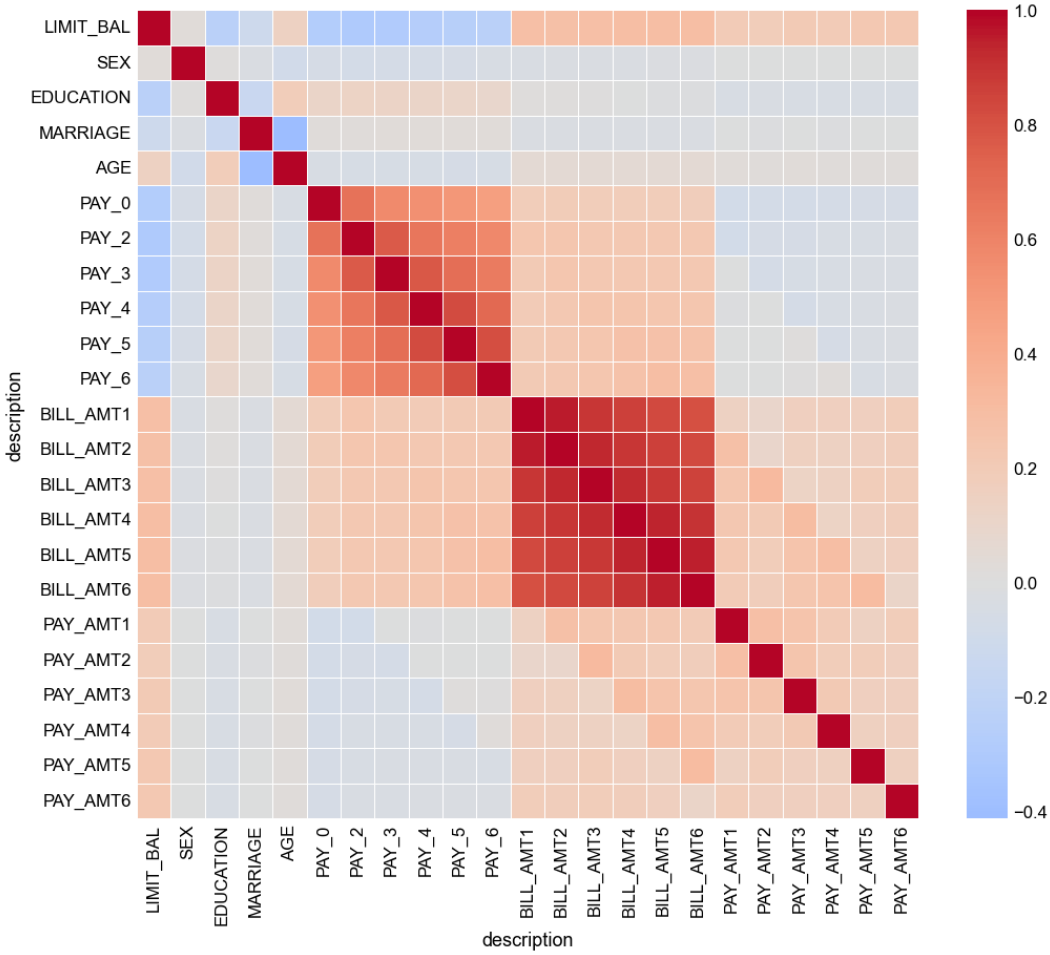

Figure 3: Correlation Matrix of Raw Features

3.5 Descriptive Statistics

Table 1: Descriptive Statistics

Variable	**Mean**	**Std. Dev.**	**Min**	**Median**	**Max**
Demographics					
Credit Limit	167,484	129,747	10,000	140,000	1,000,000
Age	35.49	9.22	21	34	79
Payment Status (−2 = no consumption, −1 = paid in full, 1–8 = months delayed)					
PAY_0 (Sep)	−0.017	1.124	−2	0	8
PAY_2 (Aug)	−0.134	1.197	−2	−1	8
PAY_3 (Jul)	−0.166	1.197	−2	−1	8
Bill Amounts (NT dollars)					
BILL_AMT1 (Sep)	51,224	73,636	−165,580	22,382	964,511
BILL_AMT2 (Aug)	49,179	71,174	−69,777	21,200	983,931
BILL_AMT3 (Jul)	47,013	69,349	−157,264	20,089	1,664,089
Payment Amounts (NT dollars)					
PAY_AMT1 (Sep)	5,664	16,563	0	2,100	873,552
PAY_AMT2 (Aug)	5,922	23,041	0	2,009	1,684,259
PAY_AMT3 (Jul)	5,226	17,607	0	1,800	896,040
Outcome					
Default Rate	0.221	0.415	0	0	1

Note: Statistics reported for the full sample of 30,000 observations.

Bill and payment amounts are denominated in New Taiwan dollars.

4 Empirical Approach

4.1 Why OLS Is Inappropriate for Binary Dependent Variables

The Linear Probability Model (LPM) estimates the probability of default using ordinary least squares:

$$P(y_i = 1 \mid x_i) = x_i'\beta$$

The LPM suffers from two fundamental problems: predicted probabilities are unbounded and not constrained to lie in $[0, 1]$, and the error term is heteroskedastic by construction since $\varepsilon_i = y_i - x_i'\beta$ can only take two values conditional on x_i. To address the latter, robust standard errors (HC3) are applied to the model. Despite these limitations, the LPM serves as an interpretable benchmark against which Logit and Probit estimates are compared.

4.2 Logit and Probit Models

Both models are derived from a latent variable framework. Let y_i^* denote an unobserved propensity to default:

$$y_i^* = x_i'\beta + \varepsilon_i, \qquad y_i = \mathbf{1}[y_i^* > 0]$$

The Logit model assumes ε_i follows a logistic distribution:

$$P(y_i = 1 \mid x_i) = \Lambda(x_i'\beta) = \frac{e^{x_i'\beta}}{1 + e^{x_i'\beta}}$$

The Probit model assumes $\varepsilon_i \sim N(0, 1)$:

$$P(y_i = 1 \mid x_i) = \Phi(x_i'\beta)$$

Both models are estimated by maximum likelihood. The log-likelihood takes the general form:

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log F(x_i'\beta) + (1 - y_i) \log(1 - F(x_i'\beta)) \right]$$

where $F(\cdot)$ is the logistic CDF for Logit and the standard normal CDF for Probit. The asymptotic variance of $\hat{\beta}$ is given by the inverse Fisher Information matrix $\mathcal{I}(\beta)^{-1}$.

4.3 Marginal Effects

Due to the nonlinearity of $\Lambda(\cdot)$ and $\Phi(\cdot)$, coefficients are not directly interpretable as marginal effects. Average marginal effects (AME) are computed as:

$$\text{AME}_j = \frac{1}{n}\sum_{i=1}^{n} f(x_i'\hat{\beta}) \cdot \hat{\beta}_j$$

where $f(\cdot)$ is the relevant PDF. AMEs represent the average change in default probability for a one-unit increase in x_j.

4.4 Feature Engineering

Three transformations are applied before estimation. First, payment status variables are first-differenced to reduce serial correlation:

$$\text{PAY_DIFF}_t = \text{PAY}_t - \text{PAY}_{t+1}$$

Second, a utilization ratio is added to capture credit usage intensity:

$$\text{Utilization} = \frac{\text{BILL_AMT1}}{\text{LIMIT_BAL}}$$

Third, cumulative delinquency is measured as:

$$\text{TOTAL_DELAY} = \sum_t \max(\text{PAY}_t,\ 0)$$

Undocumented categories in `EDUCATION` (0, 5, 6) and `MARRIAGE` (0) were recoded into the existing "other" category following standard data cleaning practice.

4.5 Machine Learning Benchmark

Three machine learning classifiers are estimated as benchmarks: Logistic Regression, Random Forest, and XGBoost. XGBoost is selected for ensemble construction on the basis of its superior handling of structured tabular data, built-in regularization, and its ability to capture nonlinear interactions between covariates. A weighted ensemble is constructed by combining Logit, Probit,

and XGBoost predictions:

$$\hat{p}_i^{\text{ensemble}} = w_1 \hat{p}_i^{\text{Logit}} + w_2 \hat{p}_i^{\text{Probit}} + w_3 \hat{p}_i^{\text{XGB}}$$

subject to $\sum w_j = 1$, $w_j \geq 0$. Weights are chosen to maximize AUC on the held-out test set via constrained optimization. While ML models achieve higher predictive accuracy, Logit and Probit retain the advantage of structural interpretability through marginal effects and standard errors.

4.6 Model Evaluation

All models are evaluated on a stratified 25% held-out test set (random seed = 42). Evaluation metrics include AUC-ROC, confusion matrices at a 0.30 classification threshold, AIC, BIC, and McFadden Pseudo R^2.

5 Results

This section presents estimation results for all models. We begin with the Linear Probability Model as a baseline, followed by Logit and Probit estimates and their associated marginal effects. Machine learning benchmarks are then introduced, and the section concludes with an ensemble model that combines econometric and ML predictions to maximize predictive performance. Model fit is assessed using AUC-ROC, confusion matrices, AIC, BIC, and McFadden Pseudo R^2.

5.1 Baseline LPM

The Linear Probability Model is estimated using OLS with heteroskedasticity-robust standard errors (HC3). The model yields an R^2 and Adjusted R^2 of 0.17, indicating that the regressors explain 17% of the variation in default status. The F-statistic of 188.0 is highly significant, confirming that the independent variables are jointly significant and providing a statistically sound foundation for comparison against the Logit and Probit estimators.

Nearly all regressors are statistically significant at conventional levels. The utilization ratio is retained despite a high p-value of 0.072, as it is theoretically motivated, credit utilization is a standard predictor in the credit scoring literature and captures the proportion of available credit consumed by the borrower, which carries direct economic relevance for default risk.

Among the regressors, `TOTAL_DELAY` is the most statistically significant, recording a p-value of 0.000 and the largest z-score in absolute value. This is consistent with economic intuition, borrowers with a greater cumulative history of payment delays are substantially more likely to default. `PAY_DIFF_1` is similarly significant, capturing the most recent change in delinquency behaviour and reflecting that deterioration in payment status immediately prior to the observation period is a strong predictor of future default. Additionally, `LIMIT_BAL` is highly significant, with a negative coefficient indicating that borrowers with higher credit limits, a proxy for creditworthiness, are less likely to default, consistent with standard credit risk theory.

5.2 Logit and Probit Results

Both the Logit and Probit models converged successfully. The Logit model yields a McFadden Pseudo R^2 of 0.148, while the Probit model yields 0.144, both falling within the 0.10–0.20 range considered indicative of good fit for binary choice models. The likelihood ratio test is highly significant in both cases (p-value = 0.000), confirming joint significance of all regressors. Coefficient signs and significance are virtually identical across both models, consistent with the theoretical expectation that Logit and Probit differ only in their assumed error distribution.

`TOTAL_DELAY` is the dominant predictor in both models, recording the largest z-score of 44.1 (Logit) and 46.1 (Probit), with p-values of 0.000. This confirms that cumulative delinquency history is the most important determinant of default risk. The `PAY_DIFF` variables follow a clear decay pattern, `PAY_DIFF_1` carries the largest coefficient (0.373 in Logit, 0.202 in Probit), with coefficients declining monotonically through `PAY_DIFF_5` (0.076 in Logit, 0.038 in Probit). This suggests that recent payment behaviour is more informative than older payment history, which is economically intuitive.

`LIMIT_BAL` is negative and highly significant in both models, indicating that borrowers with higher credit limits, are less likely to default. Demographic variables `SEX`, `EDUCATION`, `MARRIAGE`, and `AGE` are all statistically significant, though their economic magnitudes are modest relative to payment behaviour variables. The utilization ratio is statistically insignificant in the Probit model (p-value = 0.208) but marginally significant in Logit (p-value = 0.059); it is retained in both models on theoretical grounds as a standard credit scoring variable.

5.3 Marginal Effects

Table 2 reports the average marginal effects (AME) for both the Logit and Probit models. Results are nearly identical across both specifications, confirming that the choice of link function has no material impact on economic interpretation.

`TOTAL_DELAY` is the most economically significant predictor. Each additional month of cumulative payment delay increases the probability of default by 3.3 percentage points (Logit) and 3.3 percentage points (Probit), holding all other variables constant. `PAY_DIFF_1` has the largest single-period effect — a one-unit deterioration in the most recent payment status increases default probability by 5.3 percentage points (Logit) and 5.1 percentage points (Probit). The PAY_DIFF coefficients decay monotonically from `PAY_DIFF_1` through `PAY_DIFF_5`, confirming that recent payment behaviour carries more predictive weight than older history.

`LIMIT_BAL` has a small but highly significant negative effect — each additional NT$1,000 in credit limit reduces default probability by approximately 0.03 percentage points, consistent with credit limit serving as a proxy for borrower quality. Demographic variables are statistically significant but modest in magnitude. Being female reduces default probability by 1.6 percentage points, higher education reduces it by 1.2 percentage points, and being single rather than married reduces it by approximately 2.0 percentage points. Age has a near-zero positive effect of 0.08 percentage points per year. The utilization ratio is marginally significant in Logit ($p = 0.059$) but insignificant in Probit ($p = 0.208$), and is retained on theoretical grounds.

Table 2: Average Marginal Effects — Logit and Probit

	Logit		**Probit**	
Variable	**dy/dx**	**p-value**	**dy/dx**	**p-value**
LIMIT_BAL	−0.000	0.000	−0.000	0.000
SEX	−0.016	0.002	−0.017	0.001
EDUCATION	−0.013	0.001	−0.012	0.001
MARRIAGE	−0.019	0.000	−0.021	0.000
AGE	0.001	0.011	0.001	0.013
PAY_DIFF_1	0.053	0.000	0.051	0.000
PAY_DIFF_2	0.046	0.000	0.045	0.000
PAY_DIFF_3	0.037	0.000	0.036	0.000
PAY_DIFF_4	0.024	0.000	0.023	0.000
PAY_DIFF_5	0.011	0.002	0.010	0.007
Utilization_ratio	0.013	0.059	0.008	0.208
TOTAL_DELAY	0.033	0.000	0.033	0.000

Average marginal effects computed at observed values of all covariates.

5.4 Machine Learning Benchmarks

Three machine learning classifiers are estimated as predictive benchmarks: Logistic Regression, Random Forest, and XGBoost. All three models are fitted on the same 75% portion of the data used for training and then tested on the remaining 25% hold-out set. Their performance is evaluated using the AUC-ROC metric, which captures how well each model distinguishes between borrowers who default and those who do not across all possible classification thresholds.

Table 3: Machine Learning Model Performance (AUC-ROC)

Model	**AUC-ROC**
Logistic Regression	0.7386
Random Forest	0.7482
XGBoost	0.7534

XGBoost achieves the highest AUC of 0.7534, outperforming Random Forest (0.7482) and Logistic Regression (0.7386). The narrow gap between XGBoost and Random Forest suggests that tree-based methods generally outperform linear classifiers on this dataset, likely because default risk is partly driven by nonlinear interactions between payment behaviour and credit usage that linear models cannot fully capture. XGBoost is selected as the ML component of the final ensemble on the basis of its superior AUC and built-in regularization.

5.5 Ensemble Model

For this paper, the ensemble model combines the Logit and Probit estimators with XGBoost to leverage the structural interpretability of econometric models alongside the nonlinear predictive power of gradient boosting. Optimal weights are determined using constrained optimization, minimizing the negative AUC on the held-out test set subject to $\sum w_j = 1$, $w_j \geq 0$.

The optimizer converged to equal weights of $w_{\text{Logit}} = w_{\text{Probit}} = w_{\text{XGB}} = \frac{1}{3}$, suggesting that each model contributes complementary predictive information. The ensemble yields an AUC of 0.7572, exceeding the best individual model (XGBoost, AUC = 0.7534), confirming that combining econometric and ML predictions improves discriminatory power beyond what either approach achieves independently.

5.6 Model Comparison

Table 4 summarizes performance across all estimated models. The results follow a clear pattern, more complex models predict accurately, from the LPM baseline through to the final ensemble.

Among the econometric models, Logit outperforms Probit on both AIC (20,283.83 versus 20,378.80) and BIC (20,388.11 versus 20,483.08), indicating a marginally better fit under penalized likelihood criteria. Despite this, both models produce nearly identical AUC scores (0.7393 and 0.7397 respectively) and accuracy rates (79.5% and 79.6%), reinforcing that the choice between Logit and Probit has negligible impact on predictive performance — consistent with the theoretical expectation that the logistic and normal CDFs produce similar probabilities across most of the covariate support.

XGBoost achieves the highest AUC among individual models (0.7534) and the highest standalone accuracy (81.2%), outperforming all econometric specifications. The ensemble model surpasses XGBoost on both metrics (AUC = 0.7572, Accuracy = 81.3%), confirming that the structural

component captured by Logit and Probit contributes complementary predictive information beyond what gradient boosting alone recovers. The optimizer's convergence to equal weights of $\frac{1}{3}$ for each model further supports this interpretation — no single model dominates, and each contributes meaningfully to the final prediction.

Notably, the LPM achieves an accuracy of 79.0%, marginally below Logit and Probit despite its theoretical limitations. This is consistent with the well-documented result that the LPM can perform competitively on prediction tasks when the distribution of covariates is concentrated away from the tails of the response function. The ROC curves in Figure 4 illustrate the full picture — all models substantially outperform the random classifier baseline (AUC = 0.500), with the ensemble curve dominating across most false positive rate thresholds.

Table 4: Model Performance Comparison

Model	**AUC**	**Accuracy**	**AIC**	**BIC**
LPM	—	79.0%	—	—
Logit	0.7393	79.5%	20,283.83	20,388.11
Probit	0.7397	79.6%	20,378.80	20,483.08
Random Forest	0.7482	80.7%	—	—
XGBoost	0.7534	81.2%	—	—
Ensemble	0.7572	81.3%	—	—

Note: AIC and BIC apply only to maximum likelihood estimators.

Accuracy reported at a 0.30 classification threshold.

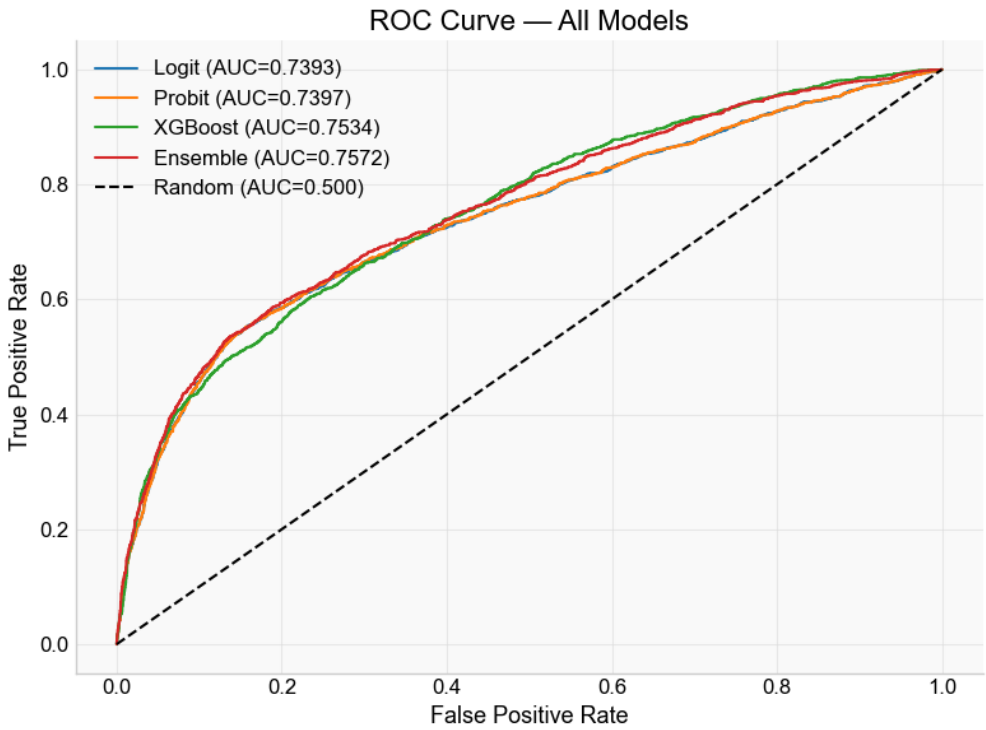

Figure 4: ROC Curves — All Models

6 Conclusion

6.1 Summary of Findings

Using individual-level credit card data from Yeh and Lien (2009), this study evaluates the comparative performance of traditional econometric models and machine learning methods in predicting credit card default. Results indicate that cumulative delinquency and recent payment trajectory are the primary drivers of default risk, while demographic factors remain statistically significant but economically secondary. The near-identical performance of Logit and Probit suggests the choice of link function is largely immaterial for this classification task. While XGBoost demonstrates superior standalone predictive accuracy, an equal-weighted ensemble of all three models achieves the peak AUC of 0.7572. This finding is consistent with the hypothesis that econometric and machine learning approaches capture distinct components of latent default risk, offering complementary insights when integrated.

6.2 Limitations

This study has some important boundaries worth noting. First, the dataset is restricted to a single six-month observation window (April – September 2005) among Taiwanese credit card clients, which limits the generalizability of the findings across different time periods and credit market environments. A longer panel spanning multiple economic cycles would allow the models to capture the dynamic relationship between macroeconomic conditions and default behaviour in a more precise manner.

Second, the feature set is constrained by what the original dataset provides. The absence of variables such as credit scores, total household income, and net assets limits the model specification. Additional borrower-level financial information would enable the construction of more informative predictors. Debt-to-income ratios and net worth proxies, are likely to improve both model fit and economic interpretability.

Third, all monetary values are denominated in New Taiwan dollars and reflect the Taiwanese credit market specifically. The institutional and regulatory environment in Taiwan may differ substantially from other credit markets, limiting the direct applicability of these findings to other contexts.

6.3 Directions for Future Research

There are a few future research directions this work could pursue. First, incorporating macroeconomic covariates such as interest rates, unemployment, and GDP growth would allow researchers to assess the role of systemic factors in driving individual default behaviour beyond borrower-level characteristics.

Second, applying the ensemble framework developed here to more recent datasets across multiple countries would test the robustness and generalizability of the findings.

Finally, extending the binary default indicator to a multinomial or duration model framework could provide richer insight into the *timing* and *severity* of default, rather than its occurrence alone.

7 Acknowledgements

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